

GENERAL PHARMACOLOGY

MBE550

UNIT 1.1.5

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LS 1.1.5- DRUG DISTRIBUTION

DISTRIBUTION

LS 1.1.5: LEARNING OBJECTIVES:

- Explain the factors that affect distribution
- Describe plasma protein drug binding and explain its pharmacokinetic implications
- Explain the concept of volume of distribution

DRUG DISTRIBUTION

INTRODUCTION:

Distribution of drugs is the process by which a drug leaves the bloodstream and enters extracellular fluids and tissues of the body

Drug distribution to the various body's tissues depends on its physicochemical properties

Drug distribution to various body tissues requires that the drug crosses several biological membranes. Distribution therefore involves the same processes as absorption, i.e. filtration, diffusion, carrier mediated transport and endocytosis.

DRUG DISTRIBUTION

INTRODUCTION ... CONT'D:

Various factors determine the rate and extent of distribution, namely lipid solubility, ionization, blood flow, and binding to plasma proteins and cellular proteins

FACTORS AFFECTING DISTRIBUTION

- Cardiac output and capillary permeability
- Tissue perfusion
- Plasma protein binding
- Tissue binding
- Placental barrier
- Blood brain barrier

FACTORS AFFECTING DISTRIBUTION....

CONT'D

Cardiac output and capillary permeability

Affect the regional blood flow and perfusion of tissues

Tissue perfusion

Well perfused tissues such as kidney, liver, heart and brain have a rapid uptake of drugs, and the drug is distributed there first. Adipose and muscle tissue are poorly perfused hence drug uptake is slower and the drug reaches these areas later

FACTORS AFFECTING DISTRIBUTION: TISSUE BINDING

- Some drugs get bound to certain tissue constituents because of special affinity for them
- Tissue binding delays elimination and thus prolongs the duration of action of the drug
- Tissue binding also serves as a reservoir of the drug

FACTORS AFFECTING DISTRIBUTION: PLACENTAL BARRIER

- The placental membrane is lipid in nature and readily allows un-ionized, lipid soluble drugs to cross the membrane. Thus some drugs taken by the mother can affect the fetus.
- Lipid soluble drugs cross the placental barrier more easily than polar drugs
- Drugs with a molecular weight of less than 600 cross the placental barrier better than larger molecules
- Drug transporters, such as the P-glycoprotein transporter, transfer drugs out of the fetus

FACTORS AFFECTING DISTRIBUTION: BLOOD BRAIN BARRIER (BBB)

- The endothelial cells of the brain capillaries lack intercellular pores and instead have tight junctions. Brain capillaries are therefore less permeable than other body capillaries. Furthermore, glial cells envelope the brain capillaries and together they constitute the BBB.
- Only lipid soluble, un-ionized drugs can cross the BBB
- The BBB may not be fully developed at the time of birth
- Inflammation such as resulting from meningitis may increase the permeability of the BBB, and thus increase the ability of ionized, poorly lipid soluble drugs to cross

PLASMA PROTEIN BINDING

Drugs bind to proteins in the plasma in varying degrees

The most important plasma proteins that can bind drugs are:

1. Albumin
2. Alpha1 acid glycoprotein
3. Lipoproteins

Acidic drugs are generally bound more extensively to albumin and basic drugs to alpha1 acid glycoprotein and/or lipoproteins

PLASMA PROTEIN BINDING CONT'D

- It is the unbound drug concentration that is more closely related to drug concentration at the active site and to drug effects
- Protein binding decreases the concentration of free drug in circulation therefore there is a limited amount of drug available to travel to the site of action
- Only free drug is able to diffuse into tissues, interact with receptors, and produce biological effects. Protein bound drugs are pharmacologically inactive.

PLASMA PROTEIN BINDING CONT'D

- It is also the free drug that is available for metabolism and excretion
- Protein binding serves as a reservoir of the drug. When the free drug levels in the plasma fall, bound drug is released. Plasma protein binding, therefore, can prolong the duration of action of the drug.

PLASMA PROTEIN BINDING: DRUG INTERACTIONS

- Many drugs may compete for the same binding sites
- Thus one drug may displace another from the binding sites resulting in a greater proportion of free drug
- This effect may increase the free drug to toxic levels
- However, there is a compensatory increase in elimination of the displaced drug since the drug released from the binding sites becomes more available for metabolism and excretion

VOLUME OF DISTRIBUTION (V_d)

- $V_d = (\text{dose} \times F) / \text{plasma concentration}$
- V_d is the proportionality constant that relates the amount of unchanged drug that reaches the systemic circulation (dose $\times$ bioavailability) to the plasma concentration achieved
- V_d is an indicator of the extent of distribution and does not represent the actual distribution of drugs in bodily fluids
- Drugs that distribute extensively have relatively large V_d values and vice versa

VOLUME OF DISTRIBUTION CONT'D

- Drugs that are lipid soluble and highly tissue bound have a high V_d and a low plasma level
- Drugs that are water soluble have a low V_d and a high plasma level
- Drugs that are highly bound on plasma proteins have low volumes of distribution while those that are extensively bound on tissues have high volumes of distribution
- V_d may vary with changes in tissue permeability and protein binding
- V_d may be influenced by age, sex, weight and disease processes (e.g. oedema, ascites)

VOLUME OF DISTRIBUTION CONT'D

- V_d enables us to estimate the plasma concentration expected for a given dose and the dose required to produce a given concentration: Plasma Conc = $\frac{\text{Dose}}{V_d}$
- However it provides little information about the specific pattern of distribution

LEARNING OUTCOMES

Students must be able to comprehend:

- Explain the factors that affect distribution
- Describe plasma protein drug binding and explain its pharmacokinetic implications
- Explain the concept of volume of distribution

QUIZ

1. Most of drugs are distributed homogeneously.

a) True

b) False

2. Biological barriers include all except:

a) Renal tubules b) Cell membranes c) Capillary walls d) Placenta

3. What is the reason of complicated penetration of some drugs through brain-blood barrier?

a) High lipid solubility of a drug b) Meningitis c) Absence of pores in the brain capillary endothelium d) High endocytosis degree in a brain capillary

4. The volume of distribution (V_d) relates:

a) Single to a daily dose of an administered drug b) An administered dose to a body weight c) An uncharged drug reaching the systemic circulation d) The amount of a drug in the body to the concentration of a drug in plasma

END

Thanks for listening